

Multiple Choice (10 marks)

1. In Python 3, what is the following function returns item from the list with max value?
 - (a) `cmp(list)`
 - (b) `len(list)`
 - (c) `max(list)`
 - (d) `min(list)`
2. Which of the following has the greatest potential for compromising a user's personal privacy?
 - (a) A group of cookies stored by the user's Web browser
 - (b) The Internet Protocol (IP) address of the user's computer
 - (c) The user's e-mail address
 - (d) The user's public key used for encryption
3. Which pillar of cybersecurity is compromised when someone logs into a system using a stolen login and password?
 - (a) Authentication
 - (b) Confidentiality
 - (c) Integrity
 - (d) Nonrepudiation
4. Let `l = [1,2,3,4]`. What is `min(l)` in Python 3?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
5. What is the value of this python expression: `1 + 3 % 3`?
 - (a) 0
 - (b) 1
 - (c) 3
 - (d) 4

True / False (10 marks)

Answer True or False

6. True or False: The function $O(N)$ grows slower than $O(N^{1/N})$ as N increases.
7. True or False: $O(1)$ represents the smallest asymptotic complexity, indicating constant time regardless of input size.
8. True or False: A program that has been tested extensively may still contain bugs that were not identified during testing.
9. True or False: A denial-of-service (DoS) attack typically targets multiple computers at once to overwhelm a network.
10. True or False: In Python 3, the expression `'a' + 'ab'` results in the string `'aab'`.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Explain how you would implement the `nextIntInRange` method to generate a random integer within a specified range. Discuss the mathematical expression needed to ensure it includes both endpoints.
12. Discuss the process of how a file is transmitted from a web server to a user after entering a web address, focusing on the role of packets and the reassembly of data.

End of Paper